A Preference-Aware Retrieval Pipeline for Cold-Start Conversational Recommendation

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Abstract

Conversational recommender systems may address the cold-start problem by eliciting user preferences through multi-turn dialogue, which makes the retrieval step unusual: the query is not given once, it is assembled incrementally as the conversation unfolds, and it must be acted upon while it is still uncertain. This extended abstract describes an ongoing project on the retrieval side of conversational recommendation. We present a training-free pipeline that turns a cold-start dialogue into a retrieval query through two mechanisms: (i) a dynamic preference centroid, which accumulates a recency-weighted representation of what the user has expressed and blends it with the current candidate set under a schedule that shifts trust toward the accumulated evidence as the dialogue proceeds; and (ii) a diversity-aware re-ranking, which balances relevance against redundancy within the set of items compatible with the user's stated constraints. We describe the full pipeline, report preliminary evidence under a target-free, recommend-only evaluation protocol, and outline the questions we are investigating.

Keywords

Conversational Recommender Systems, Cold Start, Preference Elicitation, Diversity, Evaluation.

1. Introduction

Motivation. Recommendation is particularly challenging when a platform has no information about the user to work with. New or unauthenticated users often leave minimal behavioral trace, and recommenders that rely on historical user-item interactions are ineffective in this cold-start regime [1, 2]. Conversation can be a natural strategy to apply in this context, because preferences can be elicited and refined turn by turn, and the user can observe how the system's understanding of their taste is taking shape, steer it and correct it, rather than receiving opaque suggestions from a black box. Conversational Recommender Systems (CRSs) realize this interaction model through multi-turn natural-language dialogue [3, 4]. The project presented in this paper studies a strict version of the setting, where no prior interaction history exists and the entire user model has to be built from the conversation itself.

Background. Existing CRSs fall into two families. Supervised systems learn to map dialogue to items using a knowledge graph or a pretrained language backbone, and filter the catalog through conditional constraints or entity matching [5, 6, 7]. Generative systems instead delegate the interaction to a Large Language Model (LLM) that plans the dialogue and produces recommendations by generation or by prompting over a candidate pool [8, 9]. Surveys of the area give a broader picture of how these designs differ and of the trade-offs they involve [10]. How such systems are evaluated matters as much as how they work: many protocols rely on target-aware simulators, where the interaction is implicitly driven by a hidden item to be identified, which leaks information and shifts the objective from preference elicitation to target guessing [11, 4]. We adopt a target-free protocol throughout this project.

Our Approach. In this extended abstract, we present a conversational recommender designed for strict cold start, and we focus on its retrieval side. What distinguishes retrieval in a conversational setting is that the query is never given in full: it is assembled turn by turn, and the system has to act on it while it is still uncertain. Our pipeline addresses this with two mechanisms. A dynamic preference centroid

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maintains a single point in the embedding space that summarizes what the user has expressed so far and is fed directly into retrieval, with a schedule that shifts trust toward the accumulated evidence as the dialogue proceeds. On the list this centroid produces, a diversity-aware re-ranking balances relevance against redundancy within the set of items compatible with the user's stated constraints. Both mechanisms are training-free and operate on a frozen sentence encoder and a precomputed item index. Preliminary evidence on a public dataset shows that this approach is effective.

2. The Proposed Pipeline

Overview. The pipeline processes each turn in four stages. Query understanding turns the utterance into structured constraints and into a semantic representation of what the user has expressed so far. Candidate selection narrows the catalog to a working set $\mathscr{C}_t$. The dynamic preference centroid condenses the accumulated evidence and the current candidate set into a single query vector $\mathbf{p}_t$. Diversity-aware re-ranking turns $\mathscr{C}_t$ and $\mathbf{p}_t$ into the list shown to the user. A sentence encoder maps both item representations and user utterances into the same embedding space $\mathbb{R}^d$, so the accumulated intent and the catalog are directly comparable. Item representations are built from their metadata (e.g., title, release year, genres, plot overview, keywords, tags, directors and cast for the movies domain).

Step 0: Setting Preparation. The conversation starts with no user information available to the system, and at every turn the system must return a ranked list of catalog items on the basis of the utterances observed so far. The items the user would find relevant are never revealed to the system and never used to drive the interaction. Two properties of this setting shape the design: evidence arrives incrementally, so what the user says at turn t refines rather than replaces what came before, and the system must produce a list at every turn, including the earliest ones, when its estimate of the user's needs is minimal. Keeping the relevant items hidden from the simulated user is what makes the setting diagnostic. If the simulator knew the target, it would steer the conversation toward it, and the system would be rewarded for narrowing down a hidden answer rather than for building a usable user model.

Step 1: Query Understanding. At each turn an LLM extracts a structured constraint set from the utterance. The extraction is not stateless: the model receives the constraints accumulated so far together with the dialogue history and returns their merged update (e.g., a preference expressed at turn two survives at turn ten unless the user contradicts it). Constraints are split into (i) filterable ones, which admit an exact test against catalog metadata such as release year, genres, directors or cast, and (ii) semantic ones, which include free-form descriptions and moods that no metadata field captures and that are therefore handled in the embedding space. The same step resolves film titles mentioned by the user through fuzzy matching, yielding reference items that anchor in a concrete region of the space.

Step 2: Candidate Selection. Filterable constraints are applied first. Because a conjunction of constraints elicited across several turns can easily admit no item at all, filtering is progressive rather than strict: year bounds are widened in steps when they leave the set empty, a conjunction of genres is relaxed before falling back to a disjunction, and any single constraint that empties the set on its own is abandoned rather than returning nothing. The surviving items are ranked by cosine similarity to the user representation defined below, and the top N are kept as the working set $\mathscr{C}_t$. This stage is a recall-oriented pre-filter: the ranking the user sees is decided by the two stages that follow.

Step 3: Dynamic Preference Centroid Computation. The user representation $\mathbf{u}_t$ aggregates every utterance seen so far rather than the latest one. Each utterance is encoded and the encodings are combined with an exponential recency weight $w_i \propto \gamma^{t-i}$, and the result is normalized, so recent statements dominate while earlier ones fade without disappearing. When the user has named specific films, the centroid of their embeddings is blended into $\mathbf{u}_t$, so that an explicit example pulls the representation toward the region it identifies. The query vector used for retrieval then combines this accumulated evidence with the current candidate set. Letting $\mathbf{c}_t$ be the mean embedding of the items in $\mathscr{C}_t$,

$$\mathbf{p}_t = \mathrm{norm}(\alpha_t\,\mathbf{u}_t + (1-\alpha_t)\,\mathbf{c}_t), \qquad \alpha_t = \min(\alpha_{\max},\ \alpha_0 + \delta t). \tag{1}$$

The schedule encodes an explicit trust policy. In the first turns the elicited evidence is sparse and noisy,

so the query leans on the region retrieval has already identified; as the dialogue accumulates evidence, weight shifts to the user representation and the system commits to what it has learned.

Step 4: Diversity-aware Re-ranking. Ranking $\mathscr{C}_t$ by similarity to $\mathbf{p}_t$ alone commits the entire list to the single region the current estimate points at, which is a strong bet to place on an estimate built from a handful of utterances. We therefore make redundancy an explicit term of the selection objective, so that the list hedges across the compatible region instead of collapsing onto its peak. The output list is built greedily with maximal marginal relevance [12]: starting from an empty set S, at each step we add

$$\arg\max_{i \in \mathscr{C}_t \setminus S} \quad \lambda \, \mathrm{sim}(\mathbf{p}_t, \mathbf{e}_i) \; - \; (1-\lambda) \max_{j \in S} \mathrm{sim}(\mathbf{e}_i, \mathbf{e}_j), \tag{2}$$

where $\mathbf{e}_i$ is the embedding of item i, until the list is complete. Two properties of where this step sits are worth making explicit. The re-ranking operates entirely inside $\mathscr{C}_t$, which already satisfies the user's filterable constraints and is ranked by semantic similarity to the accumulated intent, so lowering λ redistributes the list within the compatible region of the catalog and cannot import items unrelated to the request. The hyperparameter λ has a direct reading: at $\lambda = 1$ the step reduces to pure relevance ranking, and lowering it spreads the selection over the pool.

3. Preliminary Evaluation

Protocol. We evaluate on the OpenDialKG [13] dataset under the target-free, recommend-only protocol of PEPPER [11], where the simulated user is never told which items are relevant, so the interaction is driven by preference elicitation rather than by converging on an answer the simulator already knows. Each conversation runs for fifteen turns from a strict cold start over the same one hundred users, with a fixed seed and deterministic decoding, so that differences between systems are not confounded by sampling noise and every system sees the same conversations. The user simulator is Llama 3.1 8B Instruct, and we instantiate our pipeline with three open-weight backbones, so that the result is not tied to one LLM. We compare against three systems from the same benchmark suite, two supervised [5, 6] and one built on a pretrained language backbone [7], and against a prompted LLM recommender based on GPT-4o-mini, all run under identical conditions. We report preference coverage PC@k, which is the fraction of a user's relevant items that the system has surfaced within the top-k, cumulatively over the dialogue. Under this evaluation, we assume that what matters to the user is whether something useful has appeared by a given point in the conversation, not whether it appeared at one specific turn.

Results. Table 1 reports preference coverage after the full fifteen turns. Our best configuration is ahead of every baseline at every cutoff, including the prompted LLM recommender, which is the strongest of the four and relies on a substantially larger proprietary model than any backbone we use. The margin is widest at the small cutoffs that matter most in a conversational interface, where only a handful of items can realistically be shown: at $k = 10$ the best configuration roughly doubles the coverage of the

Table 1
Preference coverage (PC@k) after fifteen turns on OpenDialKG under the target-free, recommend-only protocol, with Llama 3.1 8B Instruct as the user simulator. Our pipeline is reported with each of its three CRS backbones. Best result per column in bold. Results refer to the evaluation of the pipeline described in Section 2.

Method	Conversational Backbone	PC@5	PC@10	PC@20	PC@50
KBRD [5]	Supervised	0.0050	0.0091	0.0320	0.0670
BARCOR [6]	Supervised	0.0167	0.0207	0.0498	0.0993
UniCRS [7]	Supervised	0.0233	0.0350	0.0617	0.1022
ChatGPT [11]	GPT-4o-mini	0.0287	0.0545	0.0877	0.1829
Ours	Llama 3.1 8B	0.0282	0.0563	0.0933	0.1474
Ours	Qwen3 30B	0.0301	0.0458	0.0920	0.1734
Ours	Mistral 7B	**0.0343**	**0.0827**	**0.1231**	**0.2175**

strongest supervised system. Performance varies across our own backbones, which is expected given that the same pipeline is being driven by models of very different capacity, and the ordering suggests that the pipeline benefits from a backbone that extracts constraints reliably.

4. Conclusions and Future Work

This work presented a training-free retrieval pipeline for conversational recommendation under strict cold-start conditions, where the user model is built entirely from the ongoing dialogue. The proposed approach combines a dynamic preference centroid, which progressively integrates the evidence elicited across turns, with a diversity-aware re-ranking strategy that avoids concentrating the recommendation list in a narrow region of the candidate space. Preliminary experiments under a target-free, recommend-only protocol show that the pipeline is competitive with, and in its best configuration outperforms, the considered supervised and LLM-based baselines across all evaluated cutoffs. These results provide initial evidence that explicitly modeling the gradual accumulation of preferences can be effective for retrieval in cold-start conversational recommendation.

The current study also has limitations that motivate our future work. First, the trust schedule controlling the contribution of the accumulated user representation is defined through a fixed linear function of the dialogue turn. Future work will investigate adaptive schedules driven by the amount, consistency, or stability of the evidence elicited from the user rather than by the turn index alone. Second, filterable constraints are currently handled through a progressive relaxation strategy based on hand-defined rules. We plan to study more principled mechanisms for deciding when constraints should be enforced, softened, or discarded, potentially using signals derived directly from the interaction. Third, the current evaluation focuses on recommendation effectiveness and does not yet account for the computational cost of an LLM-driven conversational pipeline. Measuring latency, inference cost, and their trade-off with recommendation quality will therefore be necessary to assess the practical deployability of the approach. More broadly, we also plan to further investigate target-free evaluation protocols, which we consider suitable for assessing whether conversational recommenders elicit and model user preferences rather than simply converge toward a target item already known to the simulator.

Declaration on Generative AI

During the preparation of this work, the authors used Claude for grammar and spelling check. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the publication's content.

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